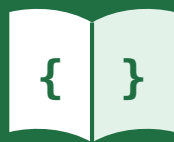


---

# Module 21 — gRPC

Exercises

Spring Boot Edu



2026-09-25

## Contents

|                                   |          |
|-----------------------------------|----------|
| <b>How to Work</b>                | <b>2</b> |
| <b>Exercise 1 — ... (Easy)</b>    | <b>2</b> |
| <b>Exercise 2 — ... (Medium)</b>  | <b>2</b> |
| <b>Exercise 3 — ... (Hard)</b>    | <b>3</b> |
| <b>Bonus Challenge (Optional)</b> | <b>3</b> |

## How to Work

1. The starter code is in `modules/21-grpc/exercise/`. Find the `TODO` comments.
2. Every exercise already has tests, and they stay **red** until you solve it.
3. Run the tests:

```
./mvnw -Pexercises -pl modules/21-grpc/exercise test
```

4. The exercise is done when all tests are green.
5. If you get stuck, read the hints first, then the solution in `modules/21-grpc/solution/`.

### ★ Tip

To run the tests of a single exercise: `-Dtest=Exercise1Test`

## Exercise 1 — ... (Easy)

**Goal:** ...

**Tasks:**

- `TODO 1` in `exercise/src/main/java/com/springbootedu/grpc/... ..`

**Hints:**

- See lesson section 3.1.

**Acceptance criteria:** all tests in `Exercise1Test` pass.

**Estimated time:** 20 minutes

## Exercise 2 — ... (Medium)

**Goal:** ...

**Tasks:**

- ...

**Hints:**

- ...

**Acceptance criteria:** all tests in `Exercise2Test` pass.

**Estimated time:** 40 minutes

## Exercise 3 — ... (Hard)

**Goal:** ...

**Tasks:**

- ...

**Hints:**

- ...

**Acceptance criteria:** all tests in Exercise3Test pass.

**Estimated time:** 60 minutes

## Bonus Challenge (Optional)

An open-ended extension idea without tests.